Dataset Description: NCR Ride Bookings

The dataset, ncr_ride_bookings.csv, contains 150,000 individual ride requests within the National Capital Region (NCR). It serves as the primary data source for building a predictive model to determine whether a ride request will ultimately be completed or result in a non-completion event (cancellation, no driver found, or incomplete).

Column Name	Data Type	Non-Null Count	Context & Role in Project
Date	Object (String)	150,000	The calendar date of the ride request. Used for temporal feature engineering (e.g., Day of Week, Holiday).
Time	Object (String)	150,000	The specific time of the ride request. Used for peak hour/time slot feature engineering.
Booking ID	Object (String)	150,000	Unique identifier for each ride request.
Booking Status	Object (String)	150,000	Primary Target Variable (Y): Indicates the final state (e.g., Completed, No Driver Found, Incomplete, Cancelled).
Customer ID	Object (String)	150,000	Unique identifier for the customer. Used for historical feature engineering (e.g., customer's past cancellation rate).
Vehicle Type	Object (String)	150,000	The requested vehicle category (e.g., Auto, Go Sedan, eBike).
Pickup Location	Object (String)	150,000	Text name of the pickup location (geographical/zone feature).
Drop Location	Object (String)	150,000	Text name of the drop location (geographical/zone feature).
Avg VTAT	Float64	139,500	Average Vehicle Travel Time to Acceptance (Avg time a vehicle takes to reach the driver when driver accepts a ride) - Potential Driver/Supply Feature.
Avg CTAT	Float64	102,000	Average Customer Travel Time to Acceptance (Avg time a driver takes to reach the customer after acceptance) - Potential Driver/Supply Feature.

Booking Value	Float64	102,000	The estimated value of the trip (Monetary Feature).
Ride Distance	Float64	102,000	The estimated distance of the trip in kilometers (Distance Feature).
Driver Ratings	Float64	93,000	Rating given by the customer to the driver (Quality Feature).
Customer Rating	Float64	93,000	Rating given by the driver to the customer (Quality Feature).
Payment Method	Object (String)	102,000	The payment type used (e.g., UPI, Cash, Debit Card).

Features Indicating Non-Completion (Missing Data Analysis)

The remaining columns primarily serve as **signals and reasons** for ride failures, which will be crucial for feature engineering. **The large number of null values in these columns is *expected*, as they only apply to non-completion events (i.e., if a ride is Completed, these columns will be null).**

Column Name	Non-Null Count	Context & Implication
Cancelled Rides by Customer	10,500	Count of rides cancelled by this customer (Historical Feature).
Reason for cancelling by Customer	10,500	Text reason for customer cancellation.
Cancelled Rides by Driver	27,000	Count of rides cancelled by this driver (Historical Feature).
Driver Cancellation Reason	27,000	Text reason for driver cancellation.
Incomplete Rides	9,000	Count of rides marked as incomplete (Historical Feature).
Incomplete Rides Reason	9,000	Text reason for incomplete status.

Target Variable Definition

The column **`Booking Status`** must be binarized for the final predictive model. The primary objective is to predict **Completed** vs. **Non-Completed** events.

- **Positive Class (Completed):** `Booking Status` = "Completed"
- **Negative Class (Non-Completed):** All other statuses, primarily including "No Driver Found," "Incomplete," and "Cancelled."